PYTORCH CAPSTONE PROJECT PROPOSAL

TITLE

Gender Classification based on Image and Generative Modeling

PROJECT DESCRIPTION

This project will develop an end-to-end pipeline for binary gender classification using facial images and explore generative modeling techniques to augment data and improve robustness. It integrates a custom CNN, fine-tuned ResNet-18, DCGAN for synthetic face generation, CycleGAN for cartoon-style transfer, and attention visualization via Grad-CAM.

WHY IT IS GOOD

Gender classification is a fundamental computer vision task with broad applications in demographics and accessibility. Combining classification, generative models, and interpretability demonstrates advanced deep learning skills and practical relevance.

HOW I WILL DO IT

1. Use a subset of 10,000 CelebA images (balanced for gender).
2. Preprocess: resize, normalize, and augment images.
3. Train a SimpleCNN; fine-tune ResNet-18 with transfer learning.
4. Implement DCGAN and CycleGAN for data augmentation and style transfer.
5. Generate additional “happy” faces with StableDiffusionPipeline.
6. Visualize model attention using Grad-CAM.
7. Develop an interface for users to upload images and compare model outputs and Meta Llama 4 Maverick analyses via OpenRouter.

DATA

• CelebA facial image dataset (subset of 10,000 images).
• Synthetic images generated by DCGAN, CycleGAN, and StableDiffusionPipeline.

EVALUATION

• Quantitative: accuracy, precision, recall, and F1-score on a held-out test set.
• Qualitative: inspection of Grad-CAM heatmaps, GAN outputs, and user-upload classification comparisons with Meta Llama 4 Maverick.

CONCLUSION AND FUTURE WORK

• This proposal outlines a comprehensive deep learning system that combines classification and generative modeling.

• Future work includes extending to multi-attribute classification, higher-resolution GAN training, and deploying a user-friendly web interface.